

In [1]:

```
!pip install seaborn
```

Requirement already satisfied: seaborn in /usr/local/lib/python3.11/dist-packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in /usr/local/lib/python3.11/dist-packages (from seaborn) (2.0.2)
Requirement already satisfied: pandas>=1.2 in /usr/local/lib/python3.11/dist-packages (from seaborn) (2.2.2)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in /usr/local/lib/python3.11/dist-packages (from seaborn) (3.10.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.11/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.3.1)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.11/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.11/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.56.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.11/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.4.8)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.11/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (24.2)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.11/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (11.1.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.11/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.2.3)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.11/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.8.2)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.11/dist-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.11/dist-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.17.0)

In [2]:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sn
```

In [3]:

```
df = pd.read_csv("heart.csv")
```

In [4]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1025 entries, 0 to 1024
Data columns (total 14 columns):
 #   Column      Non-Null Count  Dtype
---  --
 0   age         1025 non-null   int64
 1   sex         1025 non-null   int64
 2   cp          1025 non-null   int64
 3   trestbps    1025 non-null   int64
 4   chol        1025 non-null   int64
 5   fbs         1025 non-null   int64
 6   restecg     1025 non-null   int64
 7   thalach     1025 non-null   int64
 8   exang       1025 non-null   int64
 9   oldpeak     1025 non-null   float64
10   slope       1025 non-null   int64
11   ca          1025 non-null   int64
12   thal        1025 non-null   int64
13   target      1025 non-null   int64
dtypes: float64(1), int64(13)
memory usage: 112.2 KB
```

In [5]:

```
df.describe()
```

Out[5]:

	age	sex	cp	trestbps	chol	fbs	restecg
count	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000
mean	54.434146	0.695610	0.942439	131.611707	246.000000	0.149268	0.52975
std	9.072290	0.460373	1.029641	17.516718	51.59251	0.356527	0.52787
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000000
25%	48.000000	0.000000	0.000000	120.000000	211.000000	0.000000	0.000000
50%	56.000000	1.000000	1.000000	130.000000	240.000000	0.000000	1.000000
75%	61.000000	1.000000	2.000000	140.000000	275.000000	0.000000	1.000000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000000

In [6]:

```
df.head()
```

Out[6]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	52	1	0	125	212	0	1	168	0	1.0	2	2	3	0
1	53	1	0	140	203	1	0	155	1	3.1	0	0	3	0
2	70	1	0	145	174	0	1	125	1	2.6	0	0	3	0
3	61	1	0	148	203	0	1	161	0	0.0	2	1	3	0
4	62	0	0	138	294	1	1	106	0	1.9	1	3	2	0

In [7]:

```
df.target.value_counts()
```

Out[7]:

count	
target	
1	526
0	499

dtype: int64

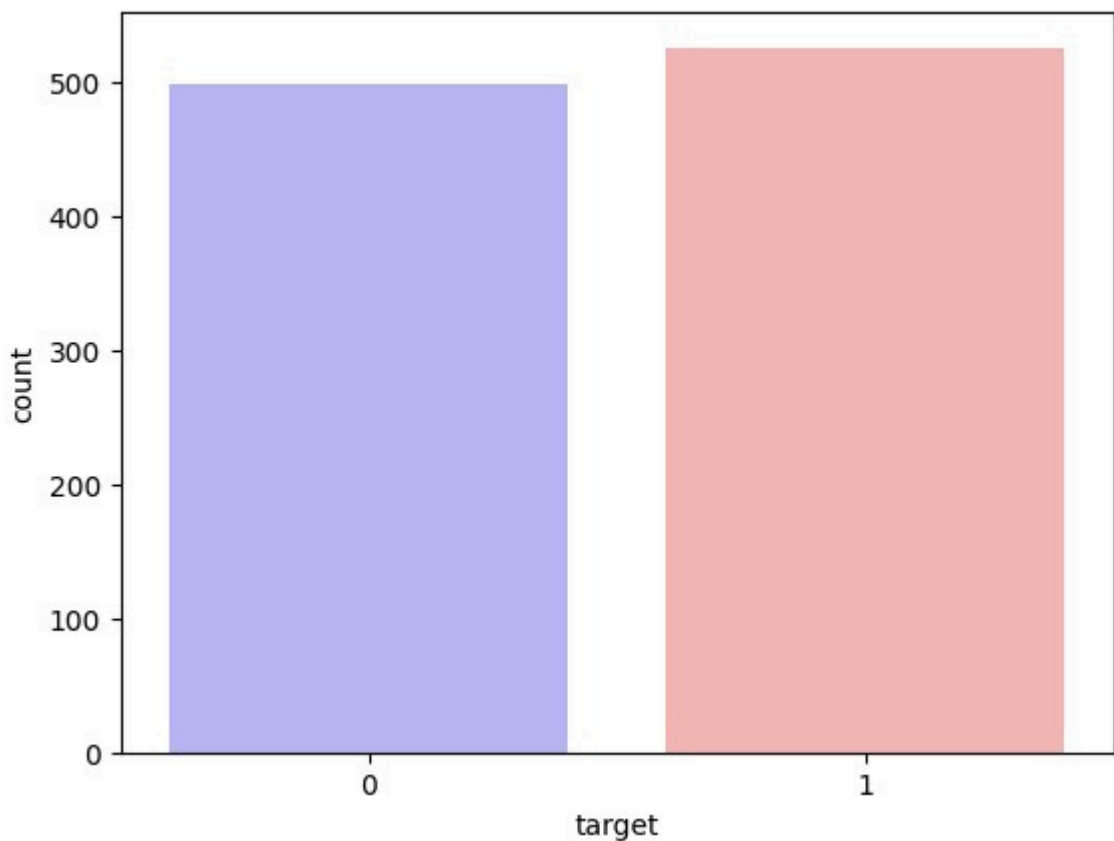
In [8]:

```
sn.countplot(x="target", data=df, palette="bwr")
plt.show()
```

<ipython-input-8-d7304cf7136c>:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sn.countplot(x="target", data=df, palette="bwr")
```



In [11]:

```
countNoDisease = len(df[df.target == 0]) countHaveDisease = len(df[df.target == 1])
print("Percentage of Patients Haven't Heart Disease: ",(countNoDisease / len(df)))
print("Percentage of Patients Have Heart Disease: ", (countHaveDisease / len(df)))
```

Percentage of Patients Haven't Heart Disease: 48.68292682926829 %
 Percentage of Patients Have Heart Disease: 51.31707317073171 %

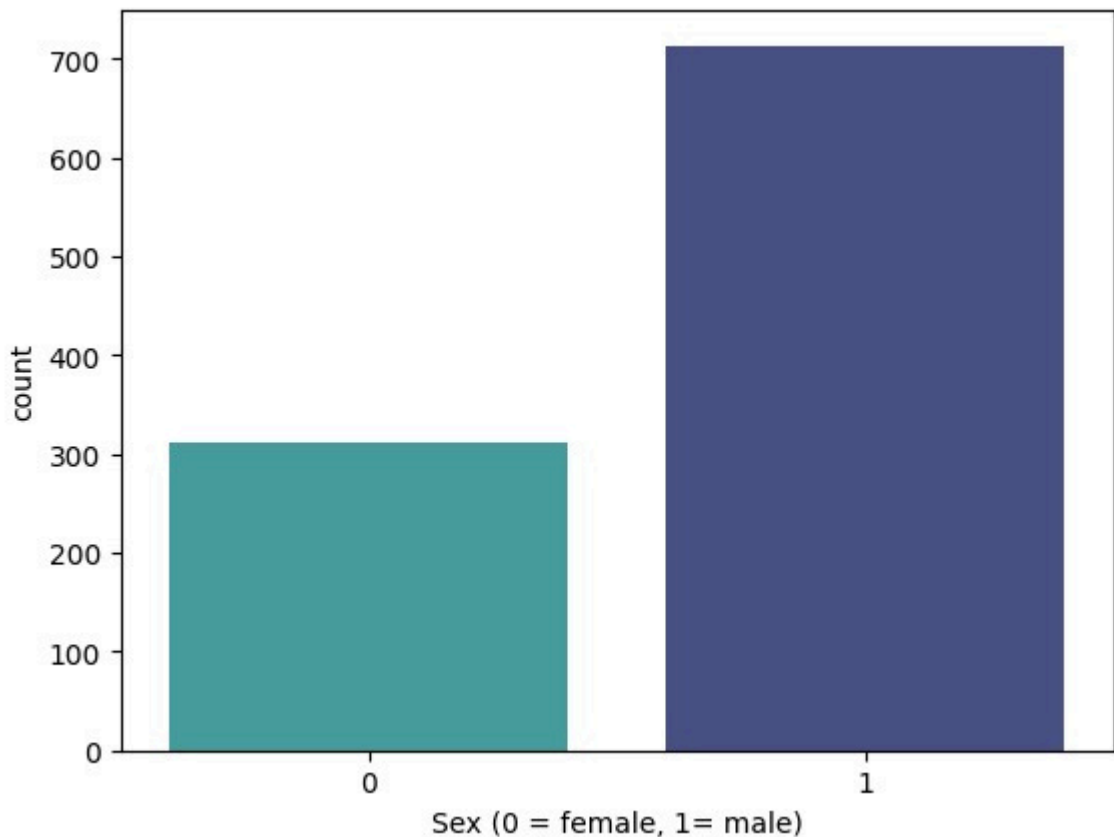
In [12]:

```
sn.countplot(x='sex', data=df, palette="mako_r")
plt.xlabel("Sex (0 = female, 1= male)")
plt.show()
```

<ipython-input-12-fc2c3fda62ec>:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sn.countplot(x='sex', data=df, palette="mako_r")
```



In [14]:

```
countFemale = len(df[df.sex == 0])
countMale = len(df[df.sex == 1])
print("Percentage of Female Patients:",(countFemale / (len(df.sex))*100), "
print("Percentage of Male Patients:",(countMale / (len(df.sex))*100), "%")
```

Percentage of Female Patients: 30.4390243902439 %
Percentage of Male Patients: 69.5609756097561 %

In [15]:

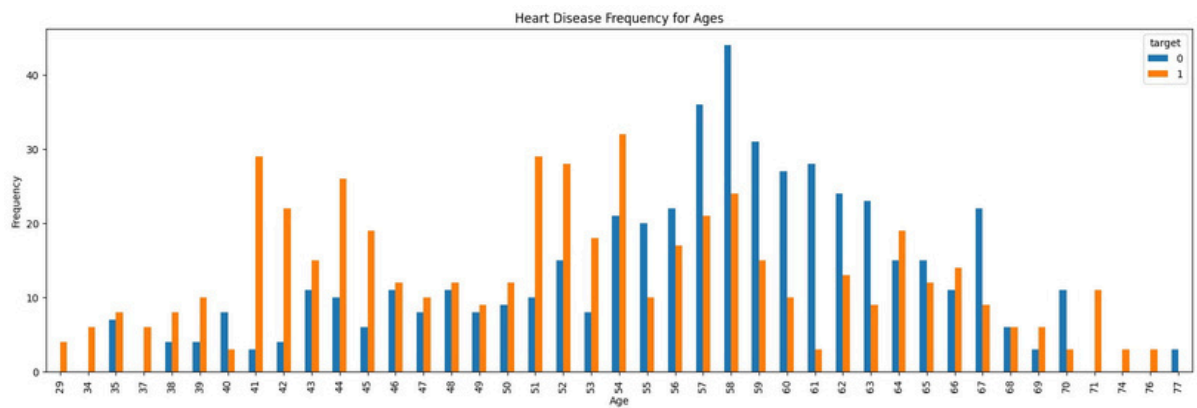
```
df.groupby('target').mean()
```

Out[15]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	target
0	56.569138	0.827655	0.482966	134.106212	251.292585	0.164329	0.456914	139.130261	0
1	52.408745	0.570342	1.378327	129.245247	240.979087	0.134981	0.598859	158.585551	0

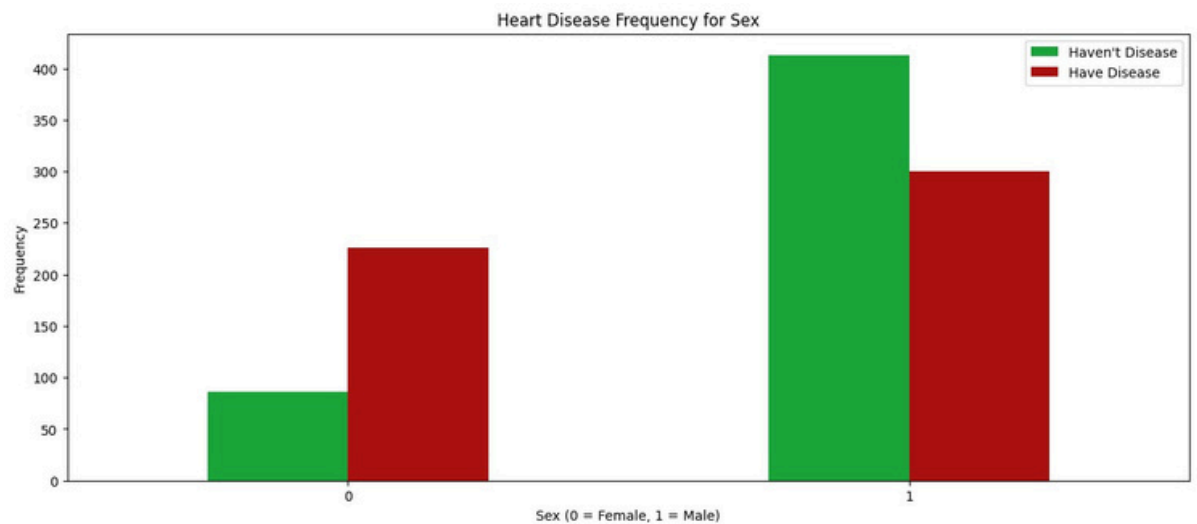
In [16]:

```
pd.crosstab(df.age,df.target).plot(kind="bar",figsize=(20,6))
plt.title('Heart Disease Frequency for Ages')
plt.xlabel('Age')
plt.ylabel('Frequency')
#plt.savefig('heartDiseaseAndAges.png')
plt.show()
```



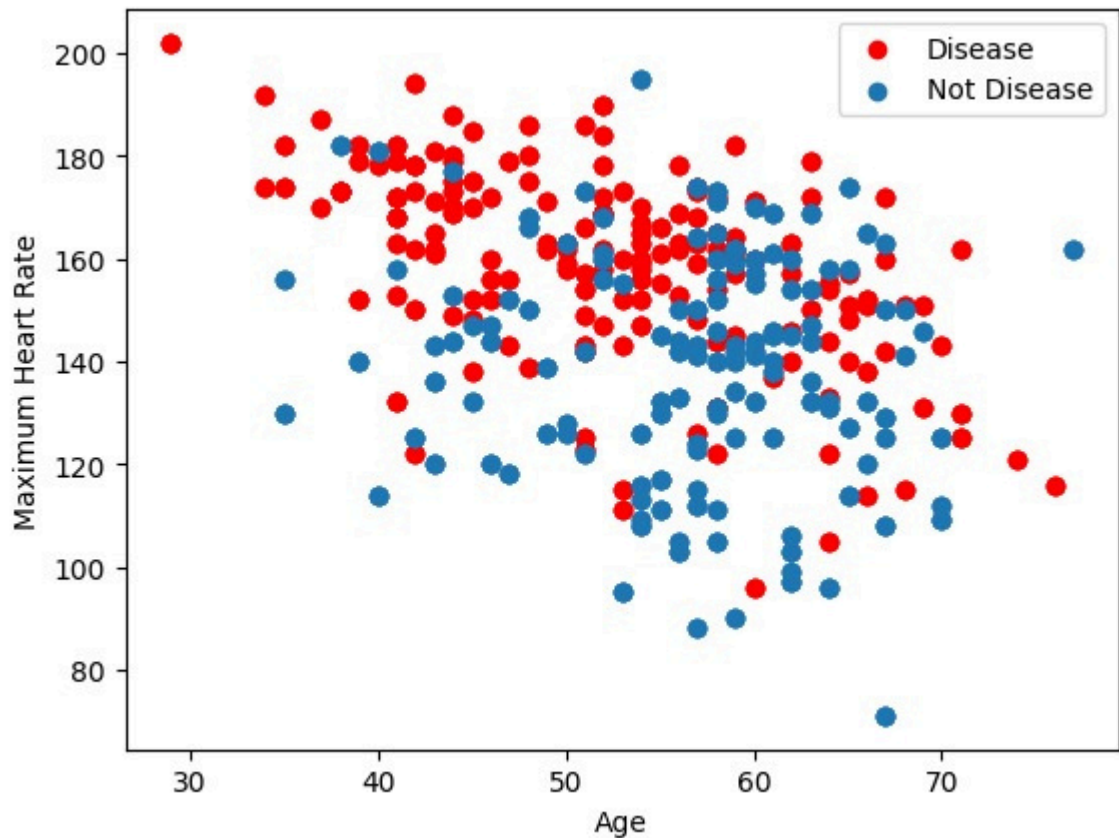
In [17]:

```
pd.crosstab(df.sex,df.target).plot(kind="bar",figsize=(15,6),color=['#1CA53',
plt.title('Heart Disease Frequency for Sex')
plt.xlabel('Sex (0 = Female, 1 = Male)')
plt.xticks(rotation=0)
plt.legend(["Haven't Disease", "Have Disease"])
plt.ylabel('Frequency')
plt.show()
```



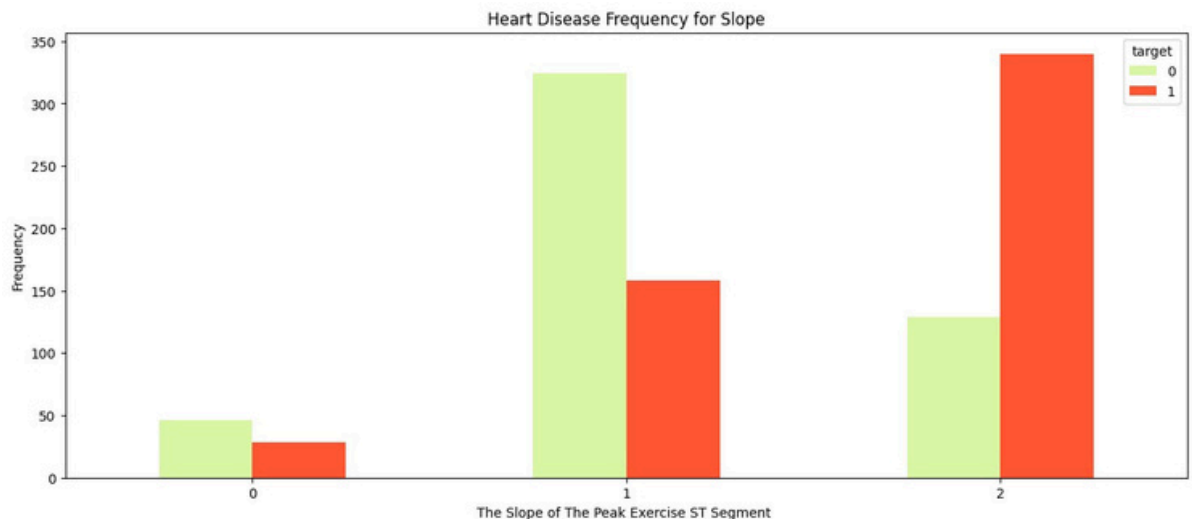
In [18]:

```
plt.scatter(x=df.age[df.target==1], y=df.thalach[(df.target==1)], c="red")
plt.scatter(x=df.age[df.target==0], y=df.thalach[(df.target==0)])
plt.legend(["Disease", "Not Disease"])
plt.xlabel("Age")
plt.ylabel("Maximum Heart Rate")
plt.show()
```



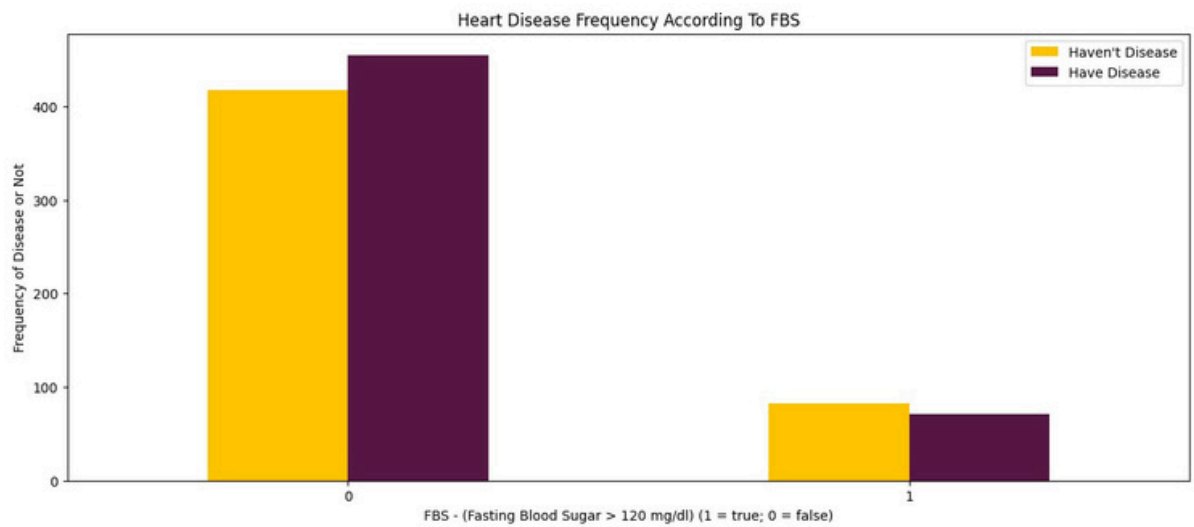
In [19]:

```
pd.crosstab(df.slope,df.target).plot(kind="bar",figsize=(15,6),color=['#DAF
plt.title('Heart Disease Frequency for Slope')
plt.xlabel('The Slope of The Peak Exercise ST Segment ')
plt.xticks(rotation = 0)
plt.ylabel('Frequency')
plt.show()
```



In [20]:

```
pd.crosstab(df.fbs,df.target).plot(kind="bar",figsize=(15,6),color=['#FFC30
plt.title('Heart Disease Frequency According To FBS')
plt.xlabel('FBS - (Fasting Blood Sugar > 120 mg/dl) (1 = true; 0 = false)')
plt.xticks(rotation = 0)
plt.legend(['Haven't Disease', 'Have Disease'])
plt.ylabel('Frequency of Disease or Not')
plt.show()
```

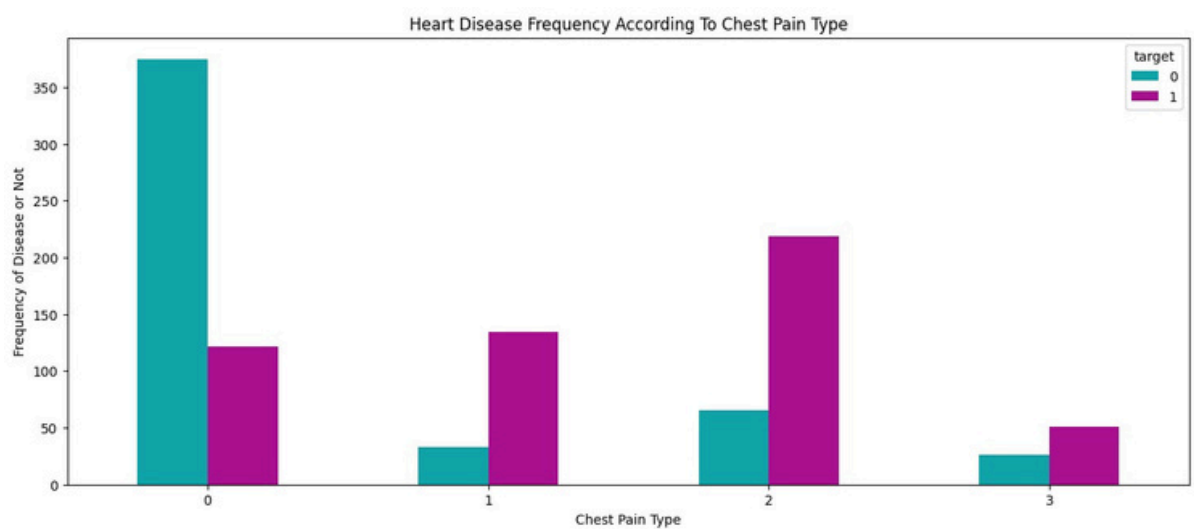


In [21]:

```
pd.crosstab(df.cp,df.target).plot(kind="bar",figsize=(15,6),color=['#11A5AA',  
plt.title('Heart Disease Frequency According To Chest Pain Type')  
plt.xlabel('Chest Pain Type')
```

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```
plt.xticks(rotation = 0)  
plt.ylabel('Frequency of Disease or Not')  
plt.show()
```



In []: